

## Exercise 6

### Date Functions

22 April 2026

Q1

```
SELECT emp_id,  
       name,  
       hire_date,  
       DATEADD(MONTH, 6, hire_date) AS hire_plus_6_months  
FROM Employees;
```

| emp_id | name    | hire_date  | hire_plus_6_months |
|--------|---------|------------|--------------------|
| 1      | Alice   | 2020-01-15 | 2020-07-15         |
| 2      | Bob     | 2021-06-10 | 2021-12-10         |
| 3      | Charlie | 2023-03-22 | 2023-09-22         |

Q2

```
SELECT student_id,  
       name,  
       DATEDIFF(DAY, dob, CURRENT_DATE) AS age_in_days  
FROM students;
```

| student_id | name   | age_in_days |
|------------|--------|-------------|
| 101        | Maya   | 7560        |
| 102        | Ethan  | 7813        |
| 103        | Sienna | 7343        |

Q3

```
SELECT event_id,  
       event_name,  
       DATEDIFF(DAY, CURRENT_DATE, event_date) AS days_remaining  
FROM Events;
```

| event_id | event_name | days_remaining |
|----------|------------|----------------|
| 1        | Seminar    | -311           |
| 2        | Workshop   | -233           |
| 3        | Hackathon  | -457           |

Q4

```
SELECT invoice_id,  
       issue_date,  
       due_date,  
       DATEDIFF(DAY, issue_date, due_date) AS days_between  
FROM Invoices;
```

| invoice_id | issue_date | due_date   | days_between |
|------------|------------|------------|--------------|
| 501        | 2025-03-10 | 2025-03-25 | 15           |
| 502        | 2025-04-01 | 2025-04-15 | 14           |
| 503        | 2025-04-10 | 2025-04-20 | 10           |

Q5

```
SELECT course_id,  
       name,  
       TO_CHAR(start_date, 'Month YYYY') AS formatted_date  
FROM Courses
```

| course_id | name       | formatted_date |
|-----------|------------|----------------|
| 201       | SQL Basics | May 2025       |
| 202       | Python     | June 2025      |

Q6

SELECT member\_id,  
DATE-FROM-PARTS (start\_year, start\_month, start\_day) AS full\_start\_date  
FROM Memberships;

| member_id | full_start_date |
|-----------|-----------------|
| 1         | 2023-05-10      |
| 2         | 2022-11-25      |

Q7

SELECT sub\_id,  
plan  
DATEADD (YEAR, 1, renewal\_date) AS extended\_renewal\_date  
FROM Subscriptions;

| sub_id | plan    | extended_renewal_date |
|--------|---------|-----------------------|
| 11     | Basic   | 2026-01-01            |
| 12     | Premium | 2026-03-15            |

Q8

SELECT order\_id,  
order\_date,  
CURRENT\_DATE AS today\_date,  
DATEDIFF (DAY, order\_date, CURRENT\_DATE) AS days\_since\_order  
FROM Orders;

| order_id | order_date | today_date | days_since_order |
|----------|------------|------------|------------------|
| 1001     | 2025-04-15 | 2026-04-22 | 372              |
| 1002     | 2025-04-10 | 2026-04-22 | 377              |

Q9

```
SELECT training_id,  
       topic,  
       EXTRACT(YEAR FROM training_date) AS training_year  
FROM   Trainings;
```

| training_id | topic      | training_year |
|-------------|------------|---------------|
| 1           | Safety     | 2025          |
| 2           | Compliance | 2025          |

Q10

```
SELECT post_id,  
       title,  
       EXTRACT(HOUR FROM published_on) AS hour_published,  
       EXTRACT(MINUTE FROM published_on) AS minute_published  
FROM   Blog_posts;
```

| post_id | title         | hour | minute |
|---------|---------------|------|--------|
| 1       | SQL TIPS      | 10   | 15     |
| 2       | Data Cleaning | 16   | 45     |

Q11

```
SELECT driver_id, license_expiry,  
       DATEDIFF(DAY, CURRENT_DATE, license_expiry) AS days_left  
FROM   Drivers;
```

| driver_id | license_expiry | days_left |
|-----------|----------------|-----------|
| 301       | 2025-08-10     | -255      |
| 302       | 2023-12-31     | -843      |

Q12

```
SELECT message_id,  
       sent_timestamp,  
       CURRENT_TIMESTAMP AS current_timestamp,  
       DATEDIFF(SECOND, sent_timestamp, CURRENT_TIMESTAMP) AS  
       seconds_since_sent  
FROM Messages;
```

| message_id | sent_timestamp      | current_timestamp   | seconds_since_sent |
|------------|---------------------|---------------------|--------------------|
| 1          | 2025-04-19 09:32:45 | 2026-04-22 00:00:00 | 31,674,435         |
| 2          | 2025-04-18 23:59:59 | 2026-04-22 00:00:00 | 31,795,201         |

Q13

```
SELECT return_id,  
       return_date,  
       DATEADD(DAY, 15, return_date) AS restock_date  
FROM Returns;
```

| return_id | return_date | restock_date |
|-----------|-------------|--------------|
| 901       | 2025-04-05  | 2025-04-20   |
| 902       | 2025-04-01  | 2025-04-16   |

Q14

```
SELECT assign_id,  
       TO_DATE(assigned_on, 'YYYY-MM-DD') AS assigned_on_date  
FROM Assignments;
```

| assign_id | assigned_on_date |
|-----------|------------------|
| 1         | 2025-03-01       |
| 2         | 2025-03-05       |

Q15

```
SELECT meeting-id,  
       TO_CHAR (scheduled-time, 'Month DD, YYYY "dt" HH:MI PM')  
       AS formatted-meeting-time  
FROM   Meetings;
```

| meeting-id | formatted-meeting-time     |
|------------|----------------------------|
| 1          | April 19, 2025 dt 02:00 PM |
| 2          | April 19, 2025 dt 09:30 AM |